

InfoBento Design Brief

UX model, requirements, and design decisions

April 2026

InfoBento Design Brief

See what matters. Skip the spiral.

Domain: Consumer electronics **Phase:** Campaign (pre-Kickstarter) **Version:** v0.9.0

1. Work Roles

Device Owner

Individual who owns an InfoBento display and places it on a counter, shelf, or windowsill.

Responsibilities:

- Configure bento boxes via web UI
- Place device on a counter or shelf with the solar panel exposed to indirect window light
- Complete first-time captive-portal Wi-Fi setup
- Update configuration occasionally as priorities change

Gifter

Person who purchases and pre-configures an InfoBento as a gift for someone else.

Responsibilities:

- Select and order device
 - Configure initial bento layout for recipient
 - Share config JSON with recipient or pre-load via captive-portal setup
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2. User Classes

Desk Decorator (Primary)

Person who values aesthetic, minimal desk or counter accessories and wants a bento-box-sized B&W eInk display that shows a rotating quote, countdown to vacation, or personal QR code – standing on its own, the body is the stand.

Goals:

- Have a visually pleasing desk or counter accessory that is also functional
- Show countdown to meaningful events
- Display a personal link or QR code for networking
- A device that charges itself near a window without any cables

Frustrations:

- Digital photo frames are too large and power-hungry
- Sticky notes with countdowns look unprofessional
- No small, elegant way to display a personal QR code
- Desk gadgets always need cables or batteries

Daily Glancer (Secondary)

Busy professional who wants a calm passive display of key daily info – weather, next event, a quote – visible at a glance from across the room. The founder is this user.

Goals:

- See today’s weather and schedule without picking up a phone
- Reduce screen time by offloading ambient info to a passive display
- Customize what info matters most to them

Frustrations:

- Checking phone for simple info leads to distraction rabbit holes
- Smart home displays are too expensive and power-hungry
- Existing e-paper gadgets are not customizable
- Every phone interaction is an attention trap

Gift Giver

Someone who wants to give a thoughtful, personalized tech gift that is useful without being overwhelming.

Goals:

- Give a unique gift that feels personal and curated
- Pre-configure the device so the recipient can use it immediately
- Gift something that does not require technical knowledge to enjoy

Frustrations:

- Most tech gifts require complex setup
- Generic gifts lack personal touch
- Subscription-based gifts feel like an obligation

3. Design Principles

Zero Device Interaction

The InfoBento itself requires no interaction – no buttons, no app to open, no charging ritual. You configure it once in a browser; after that InfoBento is just there.

InfoBento has no user-facing controls. It connects to Wi-Fi and refreshes on schedule. It charges via solar passively. The user's only interaction is glancing at it.

Do	Don't
Glance at counter display from across the room	Require button press to refresh
Solar charging near a window without cables	Dedicated companion app
Captive-portal first-time setup, no app	Manual charging schedule or cable

Pure Function Architecture

The API must be stateless – same config in, same frame buffer out, every time.

No server-side state, no user accounts, no databases. Config lives on the client. Device caches last framebuffer in flash for offline resilience.

Grayscale Elegance

Four shades – black, dark gray, light gray, white – used with intention. Every tone earns its place.

The constraint is tight enough to feel intentional: four shades, not a gradient. Think newspaper print, not a photograph.

Level	Use
Black	Important body text (conditions, entries, names)
Dark gray	Hero/display text (large, doesn't need max contrast)
Light gray	Supporting metadata (H/L, author, streaks)
White	Box interior background

Free by Default

All box types must work without accounts, API keys, or subscriptions.

All 17 box types use free data sources or local computation. Weather, forecast, sunrise/sunset, and air quality use Open-Meteo. Quotes use ZenQuotes. Everything else is pure local computation.

Web-Only Configuration

InfoBento is configured exclusively from a web browser. No native phone app required.

The web editor at infobento.com is the only configuration surface. No companion iOS or Android app for v1. This removes ~6+ months of engineering scope and the iOS background BLE risk.

Solar-Powered Counter Display

Monolithic body, no hinge, no kickstand. Body is the stand.

Enclosure ~14x11cm, white housing, minimal bezel. Solar panel on upper back. Tilt switches detect orientation for auto-rotate. Designed to survive a 4-foot drop.

4. Requirements

ID	Statement	Priority	Status
REQ-001	User can configure 3-6 bento boxes via web interface	Must	Accepted
REQ-002	Device syncs via Wi-Fi direct, no companion phone app	Must	Accepted
REQ-003	Display refreshes 1-2x/day on solar power alone	Must	Accepted
REQ-004	17 box types without user accounts	Must	Accepted
REQ-005	Web UI and API served from same port in production	Must	Accepted
REQ-006	Config persists in localStorage with JSON export/import	Must	Accepted
REQ-007	QR code gets ~half display height for scannability	Must	Accepted
REQ-008	Live PNG preview before syncing to device	Must	Accepted
REQ-009	V1 ships B&W with 4-level grayscale, no color	Must	Accepted
REQ-010	Enclosure ~14x11cm, fits GDEH0576T81 panel closely	Must	Accepted
REQ-011	Display ≥ 198 DPI, ≤ 3 mm visible bezel	Should	Accepted
REQ-012	Auto-rotate via two tilt switches, 4 orientations	Must	Accepted

5. Scenarios

Morning Glance at Kitchen Counter

User class: Daily Glancer **Setting:** Kitchen counter near window, 7:15 AM, making coffee

Sam's InfoBento is standing on the kitchen counter near the window, its solar panel catching the morning light. The crisp B&W eInk display shows weather (62F, partly cloudy), a countdown (14 days to vacation), today's date, and a motivational quote. Sam grabs a jacket based on the weather. Total interaction with the device: zero. The display refreshed on schedule at 6 AM via Wi-Fi.

Preconditions: Device completed captive-portal Wi-Fi setup; config has weather, countdown, date, and quote boxes; device refreshed on schedule.

Postcondition: User has ambient awareness of weather, date, and upcoming event without touching any device.

First-Time Web Configuration

User classes: Daily Glancer, Desk Decorator **Setting:** Home office, laptop open, just unboxed the device

Alex opens infobento.com. The web UI shows a live eInk preview on top with the editor below. They add Weather (type Portland – geocoded via Nominatim), add Countdown (anniversary date), add a Date box, and a Quote box. The live 2-bit grayscale preview updates after each change. They click Export Config and save the JSON. Later they’ll upload it during the device’s captive-portal Wi-Fi setup.

Gifting a Pre-Configured Device

User class: Gift Giver **Setting:** Living room, wrapping gifts

Jordan buys an InfoBento for their friend who loves hiking. They configure 3 boxes: weather for the friend’s city, a countdown to their next planned hike, and a QR code linking to their shared trail playlist. Jordan exports the config as JSON and includes a card with setup instructions. The friend opens the gift, connects to the device’s captive portal, enters their Wi-Fi password, and the display shows Jordan’s curated layout within a minute.

6. Wireframes

Device Display

Single B&W eInk panel, enclosure ~14x11cm (fits GDEH0576T81 panel closely with minimal bezel). 198 DPI. Tilt switches detect orientation – layout auto-rotates.

Elements:

- 3-10 sections filling the panel area (dynamic based on font size)
- Inter TTF hero font for primary data, body font for labels
- SDF-antialiased rounded box borders on grey background
- Configurable corner radius (0-10) and padding (0-10)
- 4-level grayscale: anti-aliased fonts, grey labels for hierarchy
- Multi-column layouts via horizontal splits
- Auto-rotate based on tilt switch orientation

Web Configuration Editor

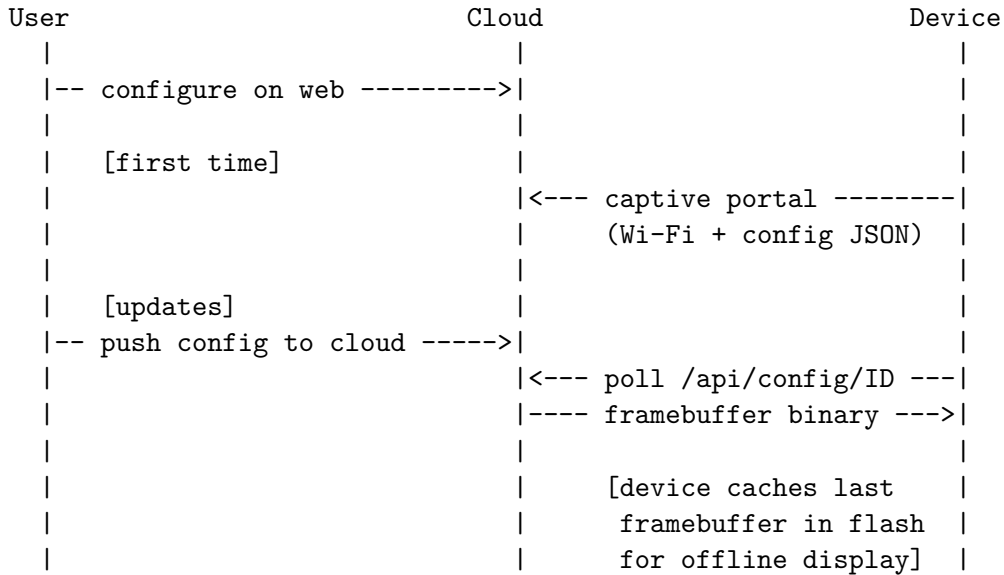
Single-column layout: preview on top, editor below. Vanilla JS with reactive state, no framework.

Elements:

- Live 2-bit grayscale eInk preview (server-rendered PNG, portrait + landscape)
- Box list with up/down reordering, remove button, inline-editable labels
- Add Box dropdown with type picker (17 types available)
- Per-box config forms (location for weather, date for countdown, URL for QR, etc.)
- Consent dialog (Privacy + Terms) on first load
- Font size, corner radius, and padding slider controls
- JSON import/export via Menu dropdown
- Browser geolocation on first load for auto-detected weather

- Debounced auto-fetch (500ms) on city input

7. Config Delivery



8. Rendering Pipeline

BentoConfig (JSON)

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Layout engine (vertical stack + horizontal splits)

- Configurable padding (0-10)
- Content-aware height allocation
- QR boxes get ~50% display height

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Box renderers (17 types)

- Inter TTF via opentype.js
- Hero (bold) + body (regular) weights
- 3-tier contrast: hero (dark grey), body (black), metadata (light grey)

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Draw primitives

- SDF-antialiased rounded rectangle borders
- Configurable corner radius (0-10)
- Grey background (GRAY_LIGHT) with white box interiors

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2-bit grayscale framebuffer (156,400 bytes)
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PNG export (4-level grayscale mapping)

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9. Findings and Decisions

Type System Fix (Fixed)

Original BentoBox interface allowed type/config mismatches (`{ type: weather, config: { type: text } }`) without type errors. Fixed with discriminated union pattern – each box type gets its own interface pairing type with config.

Display Resolution (Resolved)

Display dimensions went through multiple iterations: 240x200 (placeholder) -> 128x296 (panel-native portrait) -> 240x200 (landscape after rotation) -> 920x680 (GDEH0576T81, locked v0.7.0). Panel sourcing resolved: Good Display GDEH0576T81, 5.76", 920x680, 198 DPI.

eInk Design Research (Addressed)

Research across Tidbyt, Watchy, TRMNL, thermal printer art, and eInk phone accessories yielded key patterns:

1. One dominant element – a single large hero element taking 40-50% of height
2. Two font sizes max – large hero + small data, no medium
3. Whitespace beats borders – generous padding between sections
4. Receipt-style layout – label left-aligned, value right-aligned
5. 3-4 data items max per frame
6. Static > interactive – single-purpose static frames work best
7. No touch required – treat the display as output-only

Pivot History

- **Round 5 (Apr 21):** MagSafe clamshell with dual displays
 - **Round 10 (Apr 22):** Counter-only, single display, Wi-Fi direct
 - **Round 11 (Apr 23):** B&W (color deferred to v2), 2-bit grayscale, bento-box form factor
 - **Round 12 (Apr 24):** Panel locked (GDEH0576T81), enclosure shrunk to ~14x11cm, 17 box types, SDF borders
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10. Open Items

Item	Status	Blocking
Hardware dev kit validation (#57)	Pending order	Grey rendering quality

Item	Status	Blocking
Grey fallback – Floyd-Steinberg dithering (#56)	Deferred	Dev kit results
Wire 5 new boxes into web UI (#53)	Open	Not blocking
Content-aware layout for all types (#54)	Open	Not blocking
Price validation at volume	Needs quotes	Kickstarter pricing
SCAD enclosure model (#50)	Open	Prototype photography
Captive-portal firmware (#39)	Open	Device functionality
Tilt switch hardware (#48)	Open	Auto-rotate

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